

Probability PhD Exam, September 1989

1. Let $\{\bar{X}_m\}$ be a submartingale and T a stopping time such that $E[T] < \infty$. Assume $E[|\bar{X}_{m+1} - \bar{X}_m| | \mathcal{F}_m] \leq M$ on $(T > m)$ for some constant M . Show that $\bar{X}_T$ is integrable and $E[\bar{X}_0] \leq E[\bar{X}_T]$.
2. Let $\bar{X}_m \rightarrow \bar{X}$ in probability. Let f be continuous on $\mathbb{R}^1$. Then $f(\bar{X}_m) \rightarrow f(\bar{X})$ in probability. Prove this.
3. Can convergence a.e. be given by a metric? In other words, is there a metric ρ such that $\rho(\bar{X}_m, 0) \rightarrow 0$ is equivalent to $\bar{X}_m \rightarrow 0$ a.e.? Substantiate.
4. Explain the notion of convergence in distribution. Let $\bar{X}_m \rightarrow \bar{X}$ in distribution. Does it follow that $\bar{X}_m \rightarrow \bar{X}$ in probability? Suppose $P[\bar{X} = a] = 1$. Is it then true that $\bar{X}_m \rightarrow \bar{X}$ in probability? Substantiate all your answers.
5. Let $\bar{Y} \in L^p$, $|\bar{X}_m| \leq \bar{Y}$, and $\bar{X}_m \rightarrow \bar{X}$ in distribution. Show that $E(|\bar{X}_m|^p) \rightarrow E(|\bar{X}|^p)$.
6. Given that f is a characteristic function and $|f(1/2)| = 1$, what can you conclude?
7. State and give a sketch of the proof of the strong law of large numbers.
8. Let $\bar{X}_m$ be i.i.d., $E[\bar{X}_m] = 0$, $E[\bar{X}_m^2] = 1$. Can this sequence be complete orthonormal? Substantiate.
9. Let μ be a non-trivial infinitely divisible distribution on $\mathbb{R}^1$. Prove that μ cannot have compact support.